

Institute of Actuaries of India

May 2011 EXAMINATION

Subject ST8 — General Insurance: Pricing

INDICATIVE SOLUTION

Q.1)

The most commonly used forms of original loss curves are as follows:

First loss scales / exposure curves:

Usually seen in property business, these curves give the proportion of the full value premium allocated to primary layers limited at different values.

The curve $G(y)$ represents the expected ratio of the claim cost below a point y to the total claim cost. Here y is represented as a percentage of the maximum claim.

Example:

Let the PML for a risk = 100.

Let the total premium for the risk = 20

Now if there is a deductible of 25, and the exposure curve value at $y=0.25$ is $G(y) = 0.30$, then the premium for the cover with deductible will be $(1-0.30)*20 = 14$.

Increased Limit Factors

We choose a "basic limit"; this is usually a relatively low primary limit. We construct a table of multiplicative factors (ILFs) giving the ratio of the premium for higher limits to the basic limit premium. We usually use this terminology and format for casualty business.

Example:

Let the basic limit be 100 for which the premium is 20.

Let the ILF factor for a limit of 200 = 1.5

Then the premium for a cover with limit 200 will be $20*1.5 = 30$.

[Total Marks – 4]

Q.2)

Although GLMs are a multivariate method, there is generally benefit in reviewing some one-way and two-way analyses of the raw data prior to multivariate modeling.

In one-way analysis, the exposure, claim numbers and claim amounts are analyzed for each rating factors (and each level within the rating factor) separately.

In two way analysis the same statistics are looked at for one pair of rating factors at a time by looking at all combinations of the levels of the two rating factors.

One way analysis

The one-way analysis helps the modeler identify issues such as:

- The distribution of exposure among various levels of a rating factor. For example if more than 99% of exposures in a factor is in any one level only, it may not be useful to use this factor in the analysis.
- It also reveals the claim counts and exposure at each level of a factor. This helps in deciding on groupings of some levels so as to make the GLM analysis results more meaningful.
- In addition to investigating the exposure and claim distribution, a query of one-way statistics (for example, frequency, severity, loss ratio and pure premium) will give a preliminary indication of the effect of each factor.

Two way analysis

- A two-way analysis considers key statistics summarized by each combination of a pair of factors. It helps in identifying the correlation effects of pairs of rating variables.
- It also helps in identifying the interaction effect.
- Two way analyses also reveal the exposure and claim numbers for various combinations of levels from a pair of factors.

Correlation analysis

- It can explain why the multivariate results for a particular factor differ from the univariate results.
- It also indicates which factors may be affected by the removal or inclusion of any other factor in the GLM.
- Cramer's V statistics can be used to understand the correlation between two variables. A higher statistics implies a higher correlation.

Distribution analysis

In distribution analysis, the distribution of claim amounts is analyzed for various segments of the data.

It may help in:

- Identifying presence of large losses which need to be addresses before fitting a GLM model.
- Understanding the statistical distribution of the claim amounts
- Some other feature of the claim amounts e.g. unusual reserving patterns may also be revealed by a more detailed distribution analysis.

[Total Marks – 6]

Q.3)

- (i) As the link function used for both frequency and severity was the Log function, the relativities for the expected values can be obtained by taking exponential of the parameter estimates.

The base estimates also for both frequency and severity can be obtained by taking exponential of the intercept term.

The parameters which do not appear in the model output will have a relativity of 1.0.

The following table gives the base estimates and the relativities:

	Frequency	Severity
Base estimates	18.09%	19,930.37

Relativities:

Factor	Levels	Frequency	Severity
Age	Below 30	1.51	1.11
Age	30-65	1.00	1.00
Age	Above 65	0.75	1.05
Gender	Female	0.80	1.00
Gender	Male	1.00	1.25
Car Value	Low	1.00	0.60
Car Value	High	2.51	1.00

- (ii) The pure premium for Male, aged 35 and insuring low value car:

Expected Frequency: $18.09\% * 1.00 * 1.00 * 1.00 = 18.09\%$

Expected Severity: $19,930.37 * 1.00 * 1.25 * 0.60 = 14,913.17$

So the pure premium = 2,697.28

(iii) Business / market considerations

- The business objective of the insurer e.g. the insurer might want to increase the market share
- Competitive pressure e.g. if all other companies in the market follow a certain structure, it will be difficult to go against them.
- Difficulty in establishing a technical price e.g. where the experience is very volatile and the experience data is sparse for example vintage cars or very high end cars
- Price elasticity of demand e.g. if the new theoretically correct rates come out to be lower than the old rates but the insurer thinks that even at a price higher than the theoretical price the renewal rate will be at an acceptable level, there may not be a reason to change the rates.

- Market conditions e.g. soft or hard market, position in the insurance cycle.
- Use of no claim discounts
- Regulatory restrictions e.g. some rating factors may not be allowed as per regulation or legislation
- Sophistication of the quote and sales systems
- Training level of the sales staff
- Relationship with a particular broker or business associate e.g. auto dealers
- Data availability or other data issues
- Potential impact on a relativity factor e.g. the marketing team may be reluctant to accept a very large increase in premiums for a segment and may want to do in a gradual basis.
- Sophistication of the buyers or the distribution network.

[Total Marks – 13]

Q.4)

(i)

Desirable qualities of a complement of credibility

- accuracy as a predictor of next year's mean loss costs (low variance around next year's mean loss costs)
- unbiasedness as a predictor of next year's mean expected losses (the differences between the predictor and the subsequent loss costs should average out near zero)
- independence from the base statistic
- availability of data
- ease of computation
- Explainable relationship to the loss costs of the class or individual being rated.

(ii)

There are four types of issues to consider when choosing the complement:

Practical issues

- Availability of statistics : some statistics require more complicated programming than the others and hence some statistics may be more readily available than others. Practical considerations may hence require the actuary to choose from a statistic which is readily available
- Ease of computation : if a statistic is easy to compute it is often easier to explain to management and customers.

Competitive market issues

- If the rates are too high then the competition can under cut the rates and still make profit .
- Hence the rates have to be
- Unbiased (neither too high or too low over a large number of loss cost estimates)
- Accurate (the rate should have as low an error to variance as possible around the future expected losses being estimated)

Regulatory issues

- Inaccurate rates create a greater risk of insolvency through random inadequacies. The regulators are concerned with inadequacy because it seeks to prevent insolvencies.
- The actuary can mitigate regulatory concerns by choosing a complement that has some logical relationship to the loss costs of the class or individual being rated . It is, then, easier to explain a high rate for a class or individual in light of the related loss costs.

Statistical issues

- When mathematical models of loss are used to estimate credibility complements there may be errors both in type of model used (model error) and the specific parameters selected for the model (parameter error).
- If the complement of the credibility is accurate in its own right and relatively independent of the base statistic (which receives the credibility), the resulting rate will be more accurate.

(iii)

Claim frequency and severity are assumed to be independent.

$$\Phi(y) = 1.95 / 2 = 0.975, \text{ giving } y = 1.960$$

$$n_0 = 1.960^2 / 0.05^2 = 1536.64$$

For $X \sim \text{Gamma}(1.75, 6)$ we have $\mu_s = 1.75/6$ [½] and $\sigma_s^2 = 1.75/36$

$$n_F = n_0 + N = n_0 (1 + CV_s^2) = 1536.64 (1 + (1.75/36) / (1.75/6)^2) = 2414.72$$

using the square root rule and assuming the observed number of claims is an adequate approximation of the expected number of claims:

$$Z = (450 / 2414.72)^{0.5} = 0.432$$

[Total Marks – 15]

Q.5)

(i)

Methods for risk excess of loss pricing:

- a. **Burning cost method:** This is an experienced based method where the total cost to the layer in the past data is analyzed to estimate the expected cost to the layer in the rating period.

From the number of claims in the data, it appears that this method may be used for layer 1 and partially (blended with another method) for layer 2.

- b. **Frequency-severity method:** This is also an experienced based method where the frequency of claims and severity of claims from the data are analyzed separately to estimate the expected cost to the layer.

From the number of claims in the data, it appears that this method may be used for layer 1 and partially (blended with another method) for layer 2.

- c. **Exposure rating:** In this method. Original loss curves (exposure curves or ILF curves) are used to estimate the cost to the layer based on the exposure and premium information provided by the cedant.

This method can be use for any layer. It is more useful for the layers where there is no credible information which is layers 2, 3 and 4 in this case.

- d. **Benchmarks:** In the benchmark method, the reinsurance may have some cost benchmarks for high layers based on data submitted by all its cedant (or any other external source).

This method is also more suitable for layers with little experience which are layers 2,3 and 4 in this case.

- e. **Stochastic modeling:** In this method, stochastic simulations are used to estimate the expected cost (and other risk statistics) for the layers. The parameters used for the simulation are estimated using the experience or based on ad-hoc judgment of the actuary or the underwriter if there is no experience.

This method may also be used for all layers but estimating parameters for layers 2,3 and 4 might be difficult. Moreover, as the cover design is simple, stochastic simulation may not be required if data or resources for other methods are available.

(ii)

Advantages of frequency / severity method:

- f. Mirrors the actual underlying process. It helps in studying the number of claims and amount of claims separately.
- g. Due to separate modeling of number and amounts, it is more helpful in modeling complex reinsurance structures e.g. deductibles and limits.
- h. By separately assessing information on loss frequency and severity, we gain additional insight into aggregate loss amounts.
- i. It helps us to identify trends.

Disadvantages:

- a. Assessing the compound frequency-severity loss distribution has more onerous data requirements than assessing aggregate amounts.
- b. This approach can be time-consuming for a single risk and requires a high level of expertise.
- c. More data is required e.g. to estimate development factors and to estimate the severity distribution and parameters.
- d. It may difficult to develop individual claims to ultimate and may be a judgmental process.

(iii) Issues in trending historical frequencies and severities

- a. The frequencies and severities should be trended separately.
- b. Factors which impact the frequency trend include
 - c. Changes in risk profile
 - d. Changes in social and economic environment
 - e. Legislation changes
- f. Even though a cedant's data may provide some trends but in most cases other trends are also looked at such as benchmark trends based on a combined data of more than one 'similar' cedants or a market statistics.
- g. If the exposure measure is in monetary terms, trends should also be applied to historical exposure base.
- h. The historical frequencies should be developed to ultimate using the past trend. This will depend on the reporting delay experienced. The most recent year may be left out from the analysis.
- i. Factors which impact severity trends include
 - j. Inflation
 - k. Changes in court awards
 - l. Changes in economic conditions
- m. Changes in risk profile
- n. Trending is usually applied on the individual from ground up loss amounts, whereas the frequency trends are applied on the historical frequency at the policy level.

- o.** It may be considered to apply different trends to different sizes of claims as there may be different underlying factors behind low and high claims.
- p.** Very large losses should be treated separately on a case-to-case basis.
- q.** Choice of base periods and the period of projection should be considered carefully.

(iv) Methods to develop individual loss amounts to ultimate

- a.** We could apply an incurred development factor to each individual loss (that is, open and closed claims), reflecting its maturity, to estimate its ultimate settlement value.
- b.** A more realistic approach is to only develop open claims using “case estimate” development factors. These case estimate development factors will usually be higher than incurred development factors at the same maturity to offset the effect of not developing closed claims.
- c.** We could use stochastic development methods to allow for the variation that may occur in individual ultimate loss amounts around each of their expected values.

(v) Steps for the analysis using stochastic simulations

- a.** Collect the historical data on number and amount of claims
- b.** Apply suitable trends on frequencies and severity to bring them to the applicable time period.
- c.** Develop the frequencies and severities to ultimate using a development pattern
- d.** Fit distributions to frequency and severity
- e.** Build the reinsurance structure to filter the gross claims through the structure to calculate the net claim to the layer for each simulation
- f.** For each simulation generate number of claims and the claim amounts for each of claims.
- g.** Calculate the cost to the layer for the original term and the new proposed term.
- h.** Run a large number of such simulations and calculate the average cost to the layer for the original term and the new proposed term.
- i.** Comparing the two average costs gives the percent reduction due to cover modification

(vi) Solution table

Loss number	Gross loss amount	Net loss amount after applying the layer excess but before applying the deductible	Deductible 1		Net claim amount to Layer 1 after applying the deductibles
			Incremental	Cumulative	
1	1.10	0.10	0.10	0.10	-
2	1.25	0.25	0.25	0.35	-
3	1.50	0.50	0.50	0.85	-
4	2.00	1.00	1.00	1.85	-
5	1.20	0.20	0.15	2.00	0.05
6	1.25	0.25	-	2.00	0.25
7	2.25	1.25	-	2.00	1.25
8	3.00	2.00	-	2.00	2.00
9	1.50	0.50	-	2.00	0.45
10	2.00	1.00	-	2.00	-
Total	17.05	7.05	2.00		4.00

[Total Marks – 29]**Q.6)****(i)****Detailed investigations**

By suggesting that premium rating structure was the cause for loss it suggests that the pricing basis itself was not adequate to protect the company from adverse selection.

For example, the company may have adopted a flat rate per mille on contents irrespective of the location of the property, at a time when other companies were increasing rates for selected areas. This may have resulted in greater losses in the major cities and other urban areas.

We should investigate our profile (mix of business) against what we had planned / were expecting.

Market premium rate structure

Main investigation should start with a comparison of the company's premium basis with those of the principal competitors.

This should reveal whether our rates are:

- generally below the market
- below the market in certain specific areas, either geographically , or for low sums insured.

It will also reveal if the minimum premium is less than that of the major competitors.

If we do find that we are open to being selected against then we should compare the distribution of risks against market data.

This will highlight any areas where we seem to be attracting particularly large volumes of risks. The relative risk premiums should enable us to measure, roughly but adequately, the size of penalty we are incurring.

Claims investigation

The next investigation is to analyse claims under the contents section minimum for the past years for which data is available.

A multiway breakdown by section, peril code, sum insured and location would be desirable, but there may be too many data cells for any results to be credible.

We could produce one- or two-way analysis of claim costs for factors such as:

- section of cover
- peril code (ie type of accident giving rise to the claim)
- sum insured
- location
- distribution channel
- number of rooms.

Ideally, there should be no relation between these classifications. However, in practice there are likely to be some correlations that we should try to measure.

For example, high sums insured are more common in the city area and it may not be easy to decide whether a heavier claims experience in city (if found) is because of the location or the above average sum insured (providing more tempting targets).

We should also try to measure any interactions within the data. For example, large sums insured within certain locations could give rise to very high experience.

Next we need to calculate the overall claims ratio as the total contents claims incurred in a period over premiums earned under the contents section. This can then be compared with the claim ratio in each classification.

The premiums and claims should be scaled up to bring them into common money terms using an index relating to “contents” monetary prices when the cover applied.

We may need to adjust the claims analysis to allow for exceptional claims in some years (eg storms, floods, other severe weather).

This will enable us to see if high claim ratios are related to any particular classification or section of cover and, if so, what specific factors give rise to the adverse experience.

We could repeat the exercise, using sums insured (or number of rooms) as the unit of exposure. This will enable us to see to what extent the required risk premium rate varies between factors, and to compare this against the actual rates used by the company.

Lapse rate investigation

Investigate by each key rating factor, to see which policies are renewing and which aren't.

This should be done on a monthly basis so we can look at trends over time.

We should allow for the time lag between the renewal date and the point at which we know the policy has lapsed.

(ii)

Rating information and source

We would start with the information as deduced from our investigations in (i) above, together with the rates adopted by our main competitors.

The year to year fluctuations in most sections of a domestic household account may make it difficult to draw firm conclusions purely from our own experience, unless we combine a number of years' data. Thus, we should check whether our base data includes exceptionally high or low claims experience.

This should also be split by peril (some will be more volatile than others).

Comparison with other companies' results and market analyses for the same period will be required.

Having satisfied ourselves regarding the base claims data, we will need to load the risk premium rate for expenses and other loadings.

These will be obtained from internal office analyses. Expenses should be subdivided into fixed and variable expenses. They should also be split between those linked to numbers or amounts of claims, numbers of policies and amounts of premium.

We will need to ascertain the expected duration of the new rating series, so that we can project our base data to the appropriate period of future exposure.

In doing so we must allow suitably for inflation and trends on both claims (frequency and amount) and sums insured (because the rate is applied as ‰ of sum insured to derive the actual premium payable).

We can get an idea of inflation from:

- studies of our own claims data
- comparison with published indices
- general economic forecasts.

It is important that we do not move too far from market rates without very strong evidence.

We should also include the managements expected return on capital while keeping the rates competitive in the market.

Therefore, we need to consider the effect that our new rates are likely to have on the size of our subsequent portfolio and its risk composition and attempt to allow accordingly for any changes in our rating calculation.

(iii)

Suggestion that the rate decreases as SI increases

The important point to establish is whether or not the true risk rate (including expenses) for contents reduces with higher sums insured.

For higher sums insured, we might expect better standards of safety and security precautions to be effected by the owners than for contents of poorer properties.

Also, there is an element of fixed expenses allowed for in the premium rate, the charge for this in the quoted rate per Rs1000 sum insured should reduce.

Alternatively, properties with the larger sums insured are likely to prove a more attractive target for thieves than the lower valued properties. They are also more likely to include high valued luxury items that are easily stolen.

If this is so, then we should actually be increasing the rate for the larger sums insured, not decreasing it. However, we should also allow for the expense element described above.

The reason why the risk rate for the lowest sums insured is relatively high may be due to significant under-insurance at this level (ie sums insured are understated).

Imposing a higher premium rate for the smaller sums insured would at least mean that we got a fairer premium for these risks, even if we didn't cure the under-insurance.

A similar problem exists to some extent in the higher sum insured bands. A lower premium rate might encourage policyholders to reduce the element of under-insurance in that area.

We would probably establish a more representative structure by increasing the rate by sum insured or, better still, charge separate rates according to area.

Alternatively, sum insured itself could be used as a rating factor, or other rating factors could be used to reduce the heterogeneity.

The under-insurance problem could be solved by imposing a minimum sum insured level or a minimum premium.

A minimum premium would also help to avoid the problem of covering fixed expenses on small policies.

(iv)

Increase the policy excess

This would reduce the claims above the new excess point by a fixed amount . Claims for amounts below the new excess point would result in nil payments . Hence this would reduce the average claim amount per sum insured, especially for small claim amounts . Hence, on the face of it, the suggestion would help with losses, especially for small policies.

The key thing to look at is an analysis of the claim frequency and claim amount, with the revised level of the excess, ie what would claim payments (and hence profits or losses) have been if the

excess had been set at the higher level. This would highlight how effective the increased excess would be in reducing the losses.

This should be tested across a wide range of sums insured, to see if the increase is reasonable across the whole portfolio.

Also, try to analyse by peril. The excess may affect the perils differently.

Also, other things should still be considered:

- the impact upon sales volumes and hence total profitability if the excess is introduced
- how policyholders' behaviour might change, in particular inflating claims
- is it easy from an administrative point of view.

Finally, if the problem is under insurance, this should be addressed separately.

[Total Marks – 33]

[Total Marks – 100]
